ECG-Based Indicators of Underlying Cardiac Disease and Arrhythmia Presence

Group Information

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1. Introduction

Electrocardiograms (ECGs) serve as a fundamental tool in clinical practice for assessing cardiac function and identifying arrhythmias. Despite their widespread use, ECG interpretation often relies heavily on clinician expertise and pattern recognition, which can introduce variability and limit the ability to consistently detect subtle abnormalities. As healthcare data becomes increasingly digitized, there is growing interest in leveraging quantitative methods to support and enhance clinical decision making.

This project explores how ECG derived features can be used to better understand the presence of arrhythmia and variation across arrhythmia types. Utilizing the UCI Arrhythmia dataset, which contains a high dimensional set of ECG measurements alongside cardiologist assigned diagnoses, this study evaluates whether patterns within these measurements can distinguish between normal and abnormal cardiac rhythms. In particular, we focus on identifying which features are most strongly associated with arrhythmia and whether meaningful patterns emerge across patient groups.

Given the complexity of the dataset which is characterized by a large number of features, missing values, and mixed data types, this project emphasizes data wrangling as a critical step in preparing the data for downstream analysis. Through systematic preprocessing and validation, we aim to create a structured dataset that supports reliable analysis. While the primary focus of the project is on data preparation and quality assessment, exploratory modeling is used to assess whether the processed data can support classification of arrhythmia presence.

Overall, the goal of this project is to demonstrate how careful data wrangling and preparation can enable more effective analysis of complete clinical datasets, ultimately supporting the broader goal of improving data driven approaches for cardiovascular care.

2. Data quality assessment

We performed a quality assessment to evaluate the Arrhythmia dataset obtained from the UCI Machine Learning Repository (Guvenir et al., 1997). The dataset contains 452 patient records with 279 ECG features and a cardiologist-assigned arrhythmia classification label. The variables include demographic descriptions, such as age and sex, alongside multidimensional ECG quantitative measurements from standard 12-lead electrocardiograms. Given the large number of features and the presence of missing values, a thorough evaluation was necessary prior to visualization or further modeling to understand the relationship between the features and the normal vs arrhythmia cases. Guided by the data quality framework proposed by Weiskopf and Weng (2013), this assessment examined completeness, plausibility, concordance, and currency across three major domains: demographics, ECG features, and diagnosis labels. In addition, descriptive statistics are included to summarize the distribution, demographic characteristics, and selected ECG measurements to characterize the structure and variability of the dataset.

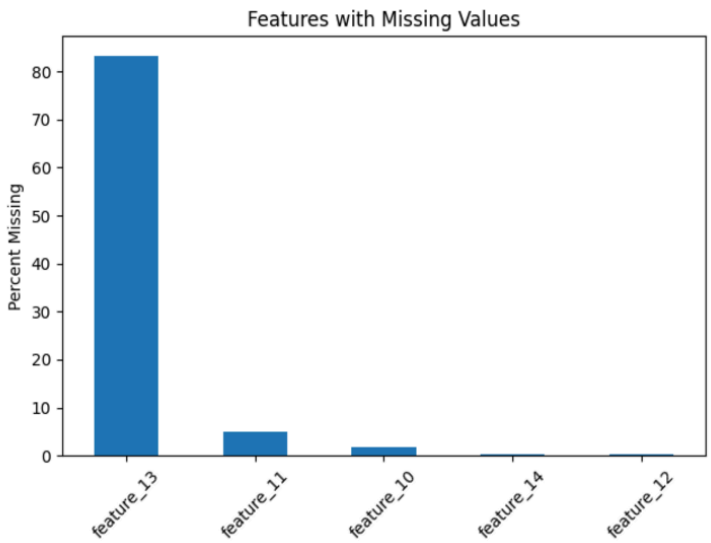

Figure 1. Percentage of missing values per feature in the UCI Arrhythmia dataset. The majority of features (279 total) have 0% missingness, but a small subset of features (10, 11, 12, 13, and 14) show substantial missing data, requiring targeted imputation strategies. Feature 14, representing the J-point value, has the highest missingness at over 80%.

Completeness was assessed by calculating the percentage of missing values for each variable. Missing variables were originally represented as "?" in the dataset, and later converted to NaN prior to analysis. While many variables demonstrated high completeness, a subset of ECG features contained substantial missingness (features 10, 11, 12, 13, and 14), requiring targeted handling strategies during preprocessing, shown in figure 1. The diagnosis distribution contains 16 classes representing normal cardiovascular rhythms and various arrhythmia subtypes. Class 1 (normal ECG) accounts for 54.2% of the dataset and constitutes the majority. The second most frequent arrhythmia class includes Class 10 (11.1%) and Class 2 (9.7%); the remaining subclasses account for fewer than 5% of observations, and a small number constitutes less than 1%. This

distribution demonstrates a class imbalance, particularly among rarer arrhythmia subtypes (Figure 2A). This imbalance may affect downstream classification performance (He & Garcia, 2009) and should be considered when interpreting the dataset and its results.

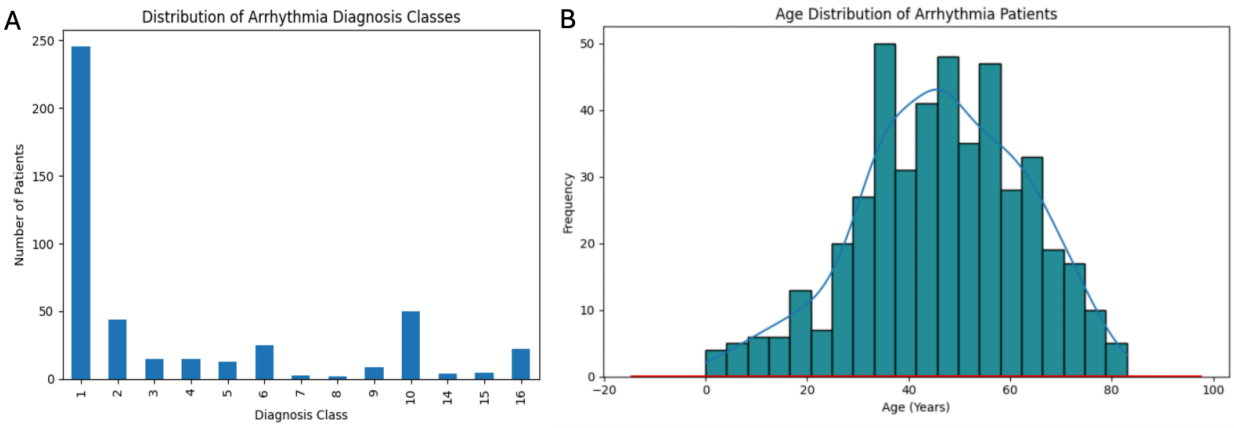

Figure 2. Arrhythmia class and patient age distribution. (A) Frequency distribution of the 16 cardiologist-assigned diagnosis classes present in the UCI Arrhythmia dataset. (B) Distribution of patient ages within the dataset.

Demographic characteristics were analyzed and age and sex variables were summarized using descriptive statistics. The mean age of the people studied in this dataset was 46.5 years (Figure 2B), with an interquartile range of 36-58 years and a maximum age of 83. The age distribution reflects a broad representation of adult patients. A minimum value of 0 years (representing newborn patients) was observed and was evaluated further during plausibility assessment. Sex variable was coded as a binary variable (0 = female, 1 = male). The sex distribution was well-balanced, with 55.1% male and 44.9% female participants. This demonstrates balanced comparisons without significant imbalance.

Beyond demographic variables, the dataset contains 275 ECG measurements representing temporal intervals, waveform durations, amplitudes and morphological characteristics extracted from the standard 12 lead ECG. These features include timing intervals such as QRS duration, PR and QT intervals (Kligfield et al., 2007), as well as amplitude measurements. Together, these variables show electrical conduction and waveform patterns that are clinically relevant for identifying arrhythmia subtypes.

Plausibility of ECG variables was assessed by reviewing summary statistics and distributional ranges across a subset of features. The ECG measurements are formatted as continuous numeric variables and show reasonable variability. While individual variables exhibit variability and include both positive and negative waveform amplitudes, no widespread structural abnormalities or significant outliers were observed across the dataset. Overall, the ECG feature set appears structurally sound and physiologically plausible for further analysis. In

addition, each of the 452 records contain a complete set of ECG measurements aligned with a corresponding diagnosis label. No duplicated observations were seen, and no missing diagnostic classifications were detected. The dataset is organized as a single patient-level table, minimizing discrepancies and supporting alignment between predictors and outcome variables.

Overall, while the dataset demonstrates reasonable structural organization and plausible ECG measurements, several challenges—including missing data, high dimensionality, and class imbalance—require careful data wrangling to ensure reliable downstream analysis. The high dimensionality of the dataset relative to the number of observations introduces additional challenges for modeling, particularly an increased risk of overfitting.

Therefore, these data quality findings directly informed downstream analytical decisions. Specifically, class imbalance across arrhythmia subtypes supported simplifying the outcome variable into a binary classification, while patterns of missing data guided imputation and feature selection strategies. Additionally, the high dimensionality of ECG features necessitated careful model selection and validation to reduce overfitting. As a result, preprocessing and modeling approaches were designed to balance interpretability with predictive performance.

3. Data Preprocessing

The goal of the preprocessing pipeline was to transform the raw UCI Arrhythmia dataset into a reliable, fully numeric, scaled feature matrix suitable for principal component analysis (PCA) and later classification. The original dataset contained 452 observations and 280 features (columns), including 279 predictors and one class label. Because the dataset contained mixed data types, missing-value indicators, high dimensionality, and several constant predictors, it required systematic cleaning before any modeling could proceed.

3.1 Raw data inspection and initial audit

Preliminary inspection of the initial records confirmed a tabular medical record format, though shape validation revealed a high-dimensional structure (452 rows by 280 columns). This ratio indicates a low sample size relative to a high feature count, requiring rigorous preprocessing before applying dimensionality reduction through PCA or classification.

The info () and dtypes checks showed a mixed structure: most columns were numeric, but five columns were stored as object type. That suggested that some entries were not clean numeric values and that the dataset had likely been imported with embedded text markers for missing data.

3.2 Data type correction and handling of inconsistent values

The object-typed columns were identified, and a sample of their unique values was printed. Those columns (11, 12, 13, 14, and 15) contained string representations of numbers, as well as the "?" symbol, the missing-value marker in the dataset (Figure 1). Before any modeling could happen, those fields had to be converted into real numeric values. Each object column was passed through pd.to_numeric(..., errors="coerce"). This process resulted in two main effects: it converted numeric-looking strings into usable numbers and turned any non-numeric tokens, including "?", into NaN. In effect, the code standardized the dataset to a single numeric format while preserving missing values for later treatment. This step is important because PCA, scaling, and most machine learning algorithms cannot operate on columns that mix text and numeric values. Converting the columns early prevented silent errors later in the pipeline.

3.3 Missing-data audit and imputation strategy

After coercion, we measured missingness across the dataset. The missing values were not evenly distributed: one variable had extremely high missingness at 83.19%, while the 4 other features had from 4.87% to only 0.22% missingness. In other words, the incompleteness was concentrated in a small group of features rather than being uniformly distributed across the whole dataset. To preserve sample size, missing values were addressed through multiple imputation to handle missing values. This approach is better than a simple mean or median fill because it estimates missing values from the relationships among the other variables. Unlike single-value imputation, which can underestimate variance, introduce bias, and distort distributions, multiple imputation better preserves the natural variability and underlying relationships in the data by generating multiple plausible values based on observed patterns, leading to more reliable, less biased estimates.

For a medical dataset with many correlated measurements, this approach helps preserve information that would otherwise be lost through a listwise deletion (complete-case analysis). The decision to impute rather than drop incomplete records was also due to the dataset already being limited to 452 observations. Removing many rows would have made the sample even smaller and could have weakened both PCA and the later classifier. Even though feature 14 had more than 80% missingness, the variable was retained through imputation due to its clinical significance. Feature 14 represents the J-point value from the ECG signal, which marks the transition between the QRS complex and the ST segment. This is clinically significant because J-point deviations are associated with various cardiac abnormalities.

3.4 Target variable handling and class structure

In the UCI arrhythmia dataset, the presence or absence of the identified cardiac arrhythmia is classified into 16 groups. Class 01 refers to 'normal' ECG classes, 02 to 15 refers to different classes of arrhythmia, and class 16 refers to the rest of the unclassified ones. The original target column was renamed from column 280 to Class to make the dataset easier to interpret. The predictors (attributes) were then split into x and y, which is standard practice before feature engineering and modeling. Class 01 accounted for 245 cases (54%), while several classes had only a few cases each. This imbalance was one of the reasons for dichotomizing the target variable (Normal versus Arrhythmia) for the downstream logistic regression analysis.

3.5 Feature reduction, zero-variance filtering, and dimensionality control

Predictors were screened for zero-variance using the standard deviation. Seventeen columns had a standard deviation of zero, and were subsequently removed. The removed features were: *144, 152, 165, 158, 20, 157, 265, 132, 133, 68, 84, 142, 205, 70, 275, 140, and 146*. This feature reduction was not merely a housekeeping step. In a dataset with 279 predictors and only 452 observations, every unnecessary column increases dimensionality and can make a model less stable. Removing constant predictors reduces computational burden, prevents noise from entering the PCA step, and slightly lowers the risk of overfitting downstream. After removing these 17 columns, the working feature set was reduced from 279 predictors to 262 informative predictors, prior to scaling.

3.6 Standardization and checks for PCA suitability

Prior to dimensionality reduction, the features were standardized to ensure that variables with larger numeric ranges did not disproportionately influence the principal components, since PCA is driven by variance. The standardized matrix was then evaluated with two common diagnostic tests, the Kaiser-Meyer-Olkin (KMO) measure and Bartlett's Test of Sphericity. For sampling adequacy, the KMO measure was 0.623, which is acceptable and indicates that the sampling and the correlation structure is suitable for grouping the variables into factors. Furthermore, the Bartlett's Test yielded a very large chi-square value and a significant p-value ($p < 0.0001$), confirming that the correlation matrix is not an identity matrix, and that the variables are significantly correlated, making them suitable for PCA. Since the goal of PCA is to compress related variables into components, running it on unrelated data is mathematically valid but practically meaningless. Therefore, the combination of scaling and diagnostic testing we

performed strengthened our methodological argument; the data were not only standardized but also checked to confirm that PCA was a reasonable next step.

3.7 Final cleaned dataset and export

After scaling, the class column was reattached to the standardized feature matrix to keep the cleaned dataset usable and later exported for supervised modeling.

4. Statistical Analysis

We analyzed the features related to heart function such as QRS, PR and QT duration histogram, in which QRS is right-skewed because higher values between 150 and 190 The data was characterized by a mean of 88 ms where a median of 85 ms (Figure 3A, C and Table 1).

Table 1. This table represents the key ECG variables in our dataset.

Variable	Obs	Mean	Std. dev.	Min	Max
qrs_ms	452	88	15.01662	54.84743	184.837
pr_ms	452	154	25.0277	67.40508	359.8638
qt_ms	452	367	35.03878	225.0961	515.814
hr_bpm	452	75	14.01551	44.18108	164.5552

QRS duration, PR interval, QT interval, Heart Rate

In our analysis the distribution of our dataset demonstrates a normal distribution of age. The mean was 46.47 years old (Figure 3B), and the standard deviation was 16.488. The youngest age (min) was 0.93 years old, while the oldest (max) was 83.04 years old, with an IQR of 22.03.

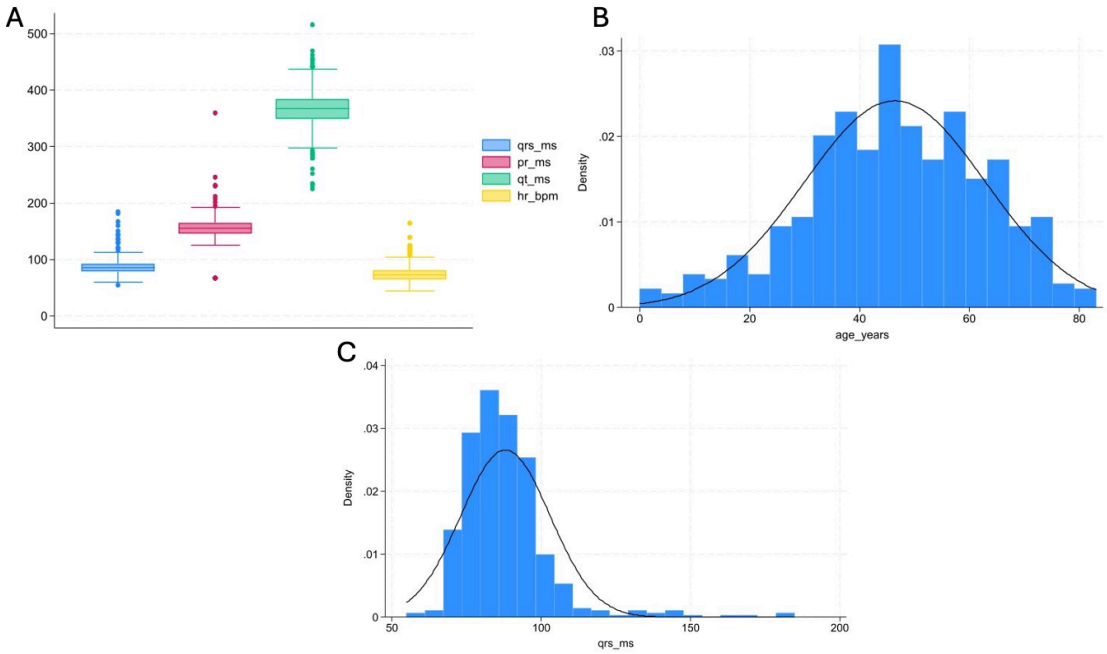

Figure 3. ECG variables and age distribution, along with density and distribution of QRS Duration. Density plot and histogram illustrating the distribution of patient age. Density plot and histogram showing the distribution of the QRS duration measurements. The distribution is characterized by a mean of 88 ms and is right-skewed.

Correlation analysis indicated that age significantly influenced both the QT interval and heart rate. The QT interval and heart rate are inversely correlated: as one increases, the other decreases (Table 2).

Table 2. Correlation Matrix of Age and Selected ECG Features.

	age_ye~s	qrs_ms	pr_ms	qt_ms	hr_bpm
age_years	1.0000				
qrs_ms	-0.0040	1.0000			
	0.9319				
pr_ms	0.0411	0.0218	1.0000		
	0.3828	0.6434			
qt_ms	0.1957*	0.2187*	0.0794	1.0000	
	0.0000	0.0000	0.0919		
hr_bpm	-0.1910*	-0.0060	-0.0447	-0.6549*	1.0000
	0.0000	0.8980	0.3433	0.0000	

Matrix displaying the Pearson correlation coefficients between Age, QRS duration, PR interval, QT interval, and Heart Rate.

5. Modeling

Next, to evaluate whether ECG-derived features could distinguish between normal and arrhythmic patterns, three classification models were used: Logistic Regression, Decision Tree, and Random Forest. “Logistic Regression is a statistical model that predicts the probability of an outcome (He & Garcia, 2009).” In our case, Logistic Regression was applied to evaluate whether a patient has a normal rhythm or an arrhythmia, based on the relationship between input variables and the outcome. This type of analysis is often used as a baseline because it is simple and easy to interpret.

Decision Tree models classify data by splitting it into branches based on feature values, creating a tree-like structure of decisions. This makes them easy to understand visually, but they can sometimes oversimplify patterns or overfit the data. On the other hand, Random Forest improves on this approach by combining many decision trees and averaging their results, which helps reduce overfitting, and improves overall stability and predictive performance.

When comparing model performance, as shown in Table 3, the Logistic Regression model achieved the highest accuracy (0.945), followed by Random Forest (0.813), while the Decision Tree model showed lower performance (0.747). The confusion matrix in Figure 4 further illustrates the performance of the logistic regression model, showing that it correctly

classified all normal cases (44/44) and most arrhythmia cases (42/47), with 5 arrhythmia cases misclassified as normal.

Table 3. Classification Model Performance Comparison

Model	Accuracy	Strength
Logistic Regression	0.945055	Strong overall performance, fewer misclassifications
Decision Tree	0.747253	Highly interpretable, shows feature relationships
Random Forest	0.813187	Balanced performance, reduces overfitting

Comparison of the predictive performance used to classify normal vs. arrhythmic ECG patterns.

Given the high dimensionality of the dataset, Principal Component Analysis (PCA) was applied as a method for reduction technique to simplify the gesture space while retaining the most important variation in the data. “PCA transforms the original variables into a smaller set of components that capture the majority of the variance (He & Garcia, 2009)”. This approach helps reduce noise and the risk of overfitting, particularly when working with datasets that contain a large number of correlated features.

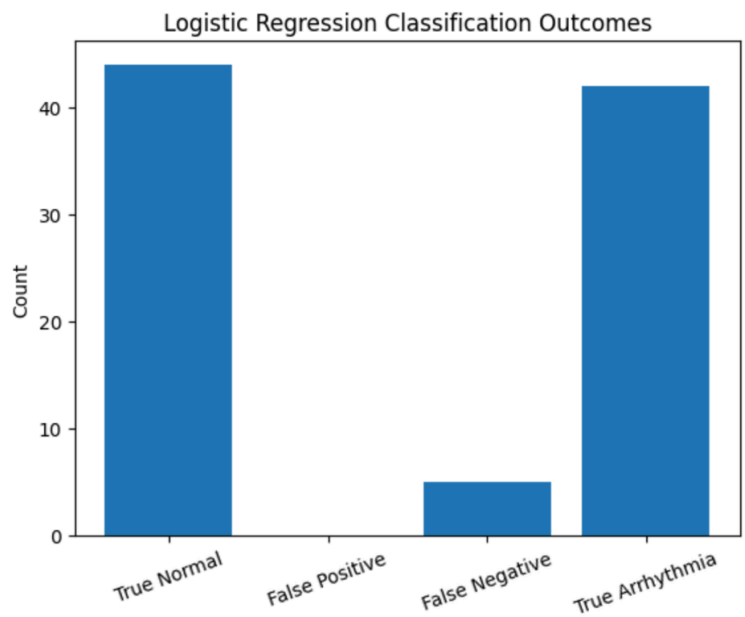

Figure 4. Confusion Matrix for Logistic Regression Model
Performance of the Logistic Regression model in classifying Normal and Arrhythmia cases.

The first two principal components were examined to assess whether meaningful separation between normal and arrhythmic cases could be observed. While some clustering patterns were present, overlap between groups indicated that the classification task remains complex, reinforcing the need for supervised models to accurately distinguish between classes.

Consistent with this complexity, Logistic Regression provided the most reliable results for this dataset, suggesting that the relationship between ECG features and arrhythmia presence can be effectively captured using a simpler, more interpretable model. While tree-based models offered additional flexibility in capturing complex patterns, they did not outperform the baseline model in this analysis.

6. Conclusion

This study successfully addressed the primary research question of utilizing the ECG features to help identify arrhythmia. By applying a rigorous data wrangling pipeline, we transformed a high-dimensional and noisy clinical dataset into a structured format suitable for analysis. Key preprocessing steps included the conversion of non-numeric markers, the use of multiple imputation for clinically significant features like the J-point (Feature 14), and the removal of zero-variance predictors, which were essential for stabilizing subsequent models.

The results demonstrated that while the classification task is complex due to class imbalance and feature overlap, a well-wrangled dataset allowed even baseline models like Logistic Regression to achieve high accuracy (0.945) in distinguishing normal rhythms from arrhythmias. Ultimately, this project highlights that the reliability of clinical predictive modeling is directly dependent on the systematic preparation and quality assessment of the underlying data.

References

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